INFO SHEET: Discrete Input Measurement Devices Report

WRITE A REPORT (You will upload this report in UNIT 3)

- Use Kuphaldt Chapter 9 for this report and use the web-links I furnished you below (click on the hyperlinks in the first column, then look at the webpage. *The web-links are in column one. click them. They are hotlinks!*
- Create a MS Word document table as shown below. *This means, open Microsoft Word, then insert a table that looks like my example.*
- Entitle the document "Discrete Input Measurement Devices" *and place the report name at the top of the report.*
- Copy the first column below, then paste it as your table's first column. *Then, use the web-link to learn more about each device*
- In the second column, state your definition of the switch type. Create at least three sentences which define the term in your own words. *Write your understanding clearly here with full sentences. USE YOUR OWN WORDS or you will get zero credit. Learn to put your thoughts on paper. Plagiarism is checked automatically. It checks copying other student ideas as well as the internet. You will be caught unless you use your own words. Plagiarism checks are done by Blackboard automatically.*
- In the third column, paste an image that illustrates how the switch works. *Look at my example below to realize what to do.*
- In the fourth column, give an application for the switch type. I require at least three full sentences which clearly evidences your thinking about how the switch is applied (used). Do not copy text from anywhere. This is to be in your own words. It is really easy to catch plagiarizers. *Write your understanding clearly here with full sentences. USE YOUR OWN WORDS or you will get zero credit. Learn to put your thoughts on paper. Plagiarism is checked automatically. It checks copying other student ideas as well as the internet. You will be caught unless you use your own words. Plagiarism checks are done by Blackboard automatically.*
- In the fifth column, ***place an image of an actual device from a manufacturer's web site. State the manufacturer, model number, and the price***.

NOTES:

- Do not copy anything word for word. Read about the switch in Kuphaldt, study the web-link, then then restate in your own words. I use a plagiarism checker, Safe Assign, to discover if you copy and paste text.
- Do college level work with professionalism in mind
- Time/effort/completeness matters

WEB-LINK	YOUR DEFINITION	PRINCIPLE OF OPERATION	INDUSTRIAL APPLICATION	EXAMPLE FOR PURCHASE
• hand switch (https://www.iqsdirectory.com/articles/electric-switch/push-button-switches.html)				

• <u>**limit switch (https://instrumentationtools.com/limit-switch-works/)**</u>				

• proximity switch (https://www.lunchboxsessions.com/materials/position-sensors/proximity-sensor-basics-lesson)				

• temperature switch (https://www.lunchboxsessions.com/materials/temperature-sensors/temperature-switches-lesson)				

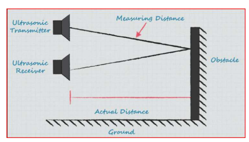
• float level switch (https://www.iqsdirectory.com/articles/level-switch.html)				

• **pressure switch (https://www.iqsdirectory.com/pressure-switches/)**				

• differential pressure switch (https://www.iqsdirectory.com/articles/pressure-switch/differential-pressure-switch.html)				

• **[flow switch (https://www.iqsdirectory.com/articles/flow-meter/flow-switches.html)]**				

Below is an example for you to follow. HINT: I used Windows Snip & Sketch tp capture the images, then I saved the image to the desktop so I could find it. Then I pasted the image in the table. I grabbed a corner of the image to make it smaller.

• ultrasonic switch (https://www.allaboutcircuits.com/industry-articles/understanding-the-basics-of-ultrasonic-proximity-sensors/)	An ultrasonic switch is a device that senses objects by first directing ultrasound in a certain direction, then it detects the the ultrasound that returns to the switch when an object interrupts the pathway.		Many industrial processes store materials that are hazardous to humans in tanks. Instead of installing a float switch which would get contaminated easily, I would use this technology because only sound touches the process. I would mount the switch vertically so it could sense a rising level, calibrating it to switch when the level reached a certain height. When a technician needs to replace it, there is little danger the tech will touch the dangerous material.	FLOWLINE - EchoSwitch® LU74-78 Ultrasonic Level Switch & Controller  LU74-5004 5.5m (18′) range, 95-250 VAC, 2″ NPT $1,100.00